CS-471: Machine Learning

Week 1

Instructor: Dr. Hashir Kiani

Classroom Etiquette and Participation Guidelines

- Treat everyone with respect and courtesy at all times
- Speak only when it is your turn and raise your hand before contributing to the discussion
- Do not interrupt others while they are speaking
- Join the class on time
- Keep mobile phones on silent during the lecture

About me

- Bachelors in Electrical Engineering from MCS, NUST (2011)
- Masters in Communications Engineering and Networks from University of Birmingham, UK (2013)
- PhD in Computer Science from University of Manchester, UK (2020)
- Research interests are applications of Machine Learning and Deep Learning in different domains including energy, healthcare, agriculture and climate change.
- Contact Information:
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  - Office: A-104, Ground Floor, Faculty Block, SEECS

Assessments

Assessment		Percentage
Theory: 75%		
1	Quizzes	10%
2	Assignments	5%
3	Project	10%
4	Mid-Term Exam	30%
5	End-Semester Exam	45%
Labs:25%		
1	Lab Work	60%
2	Final Lab	20%
3	Project	20%
Total:		100%

Assessments

- Approximately three hours of lectures every week.
- Lecture slides will be shared on LMS. **(Enrolment Code: 923870654)**
- Assignments given and submitted on LMS. Late submission penalty of 10% per late day.
- Your writings must be your own thoughts.
- Cheating and plagiarism will not be tolerated and will lead to strict penalties.

Assignments

- **Assignment 1: Week 4**
  - Given on: 9th February 2026
  - Due Date: 22nd February 2026

- **Assignment 2: Week 10**
  - Given on: 23rd March 2026
  - Due Date: 5th April 2026

- **Assignment 3: Week 12**
  - Given on: 6th April 2026
  - Due Date: 19th April 2026

Quizzes

- **Quiz 1**: 3rd February 2026 (Week 3)
- **Quiz 2**: 19th February 2026 (Week 5)
- **Quiz 3**: 3rd March 2026 (Week 7)
- **Quiz 4**: 2nd April 2026 (Week 11)
- **Quiz 5**: 16th April 2026 (Week 13)
- **Quiz 6**: 30th April 2026 (Week 15)

Project Timeline

- Project will be given on 2nd February 2026 (Week 3).
- You must get the Project proposal approved by 22nd February 2026 (Week 5).
- Project Report must be submitted by 3rd May 2026 (Week 15).
- Project Presentations will be taken in the last week tentatively.

Course Learning Outcomes

- At the end of the course, the students will be able to:

1. **Explain** theoretical concepts behind various machine learning algorithms, tools and techniques.
2. **Analyze** a given problem and select the most suitable machine learning technique for the task.
3. **Develop** solutions for practical challenges by employing machine learning algorithms.
4. **Implement** various machine learning algorithms using modern software libraries for a range of applications.

Course Prerequisites

- Linear Algebra
- Probability
- Calculus
- Programming in Python

Course Contents

- Linear Regression
- Gradient Descent
- Logistic Regression
- Decision Trees
- K Nearest Neighbours
- Parzen Window Classifier
- Support Vector Machines
- Debugging ML and Error Analysis
- Generative reasoning algorithms
- Model Selection and Experimental Design
- Clustering Algorithms
- Dimensionality Reduction
- Neural Networks
- Reinforcement Learning
- ML Models Deployment

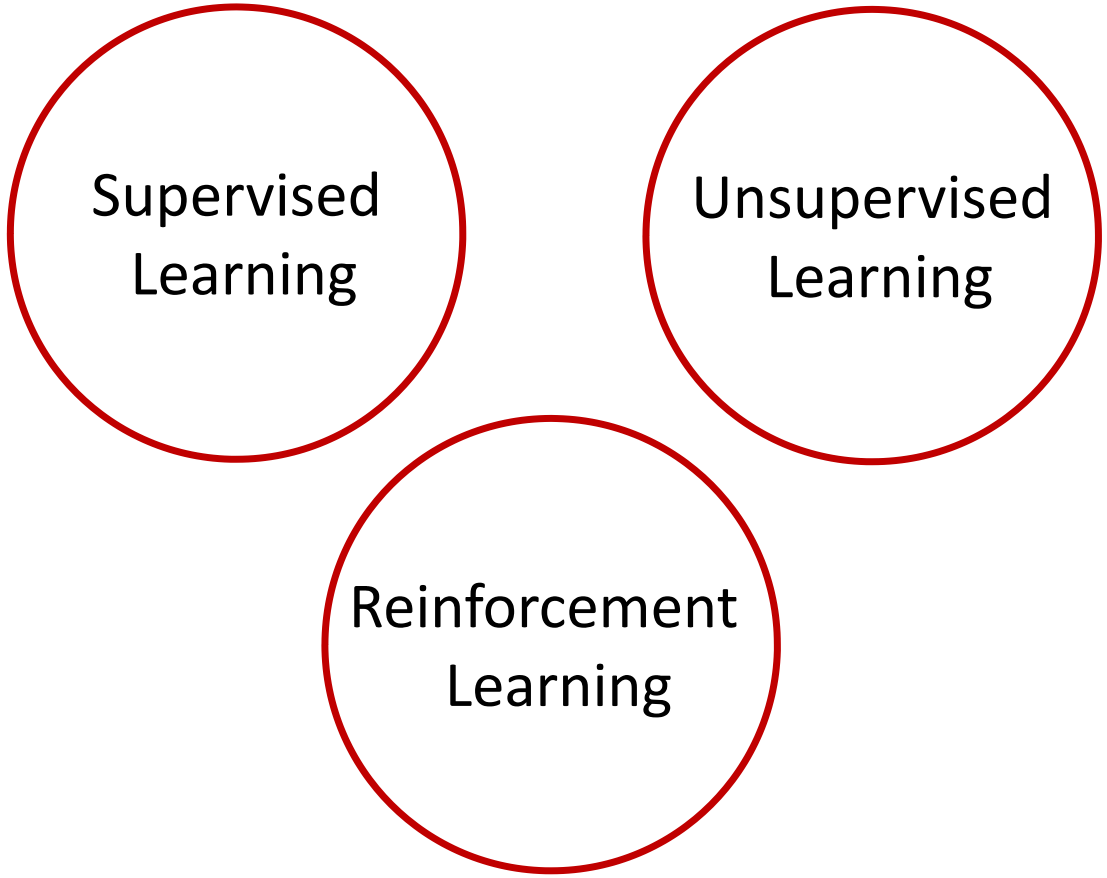
Recommended Books

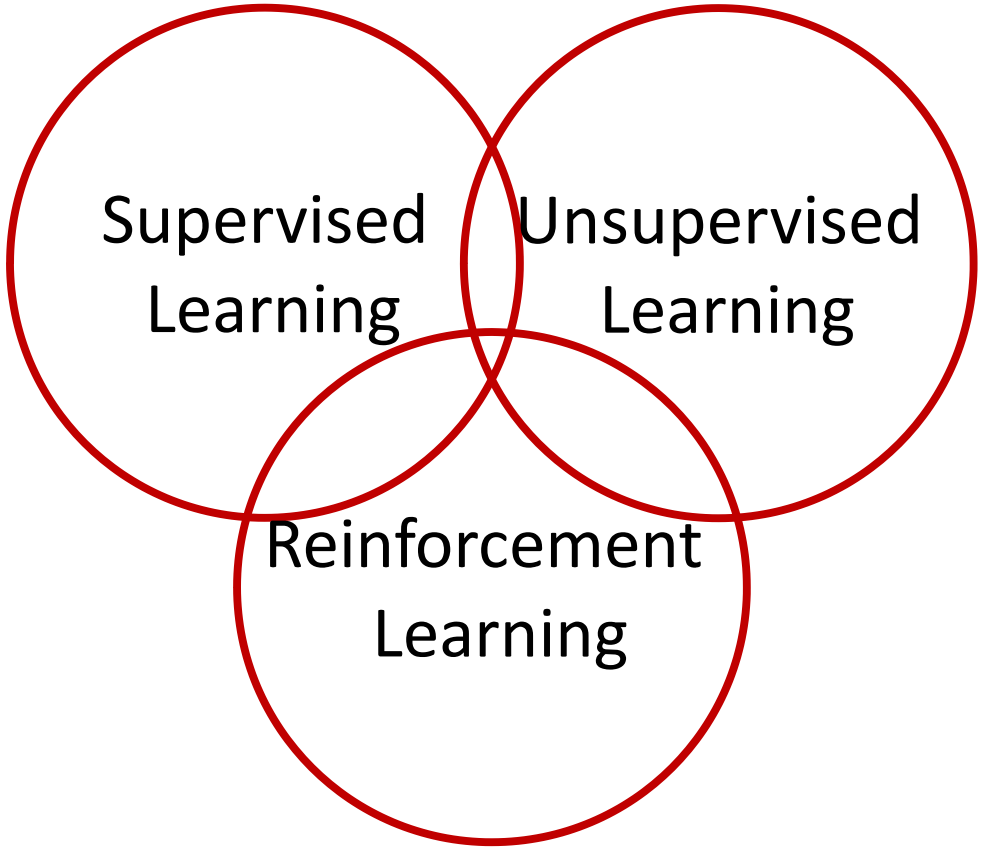
- Textbook:
  - RS Michalski, JG Carbonell, TM Mitchell, “Machine Learning: An Artificial Intelligence Approach”, Springer, 2014
  - Christopher M. Bishop, “Pattern Recognition and Machine Learning”, 2007
- Reference Books/Materials:
  - Richard Duda, Peter Hart and David Stork, Pattern Classification, 2nd ed. John Wiley & Sons, 2001.
  - Richard Sutton and Andrew Barto, Reinforcement Learning: An introduction. MIT Press, 1998

Machine Learning

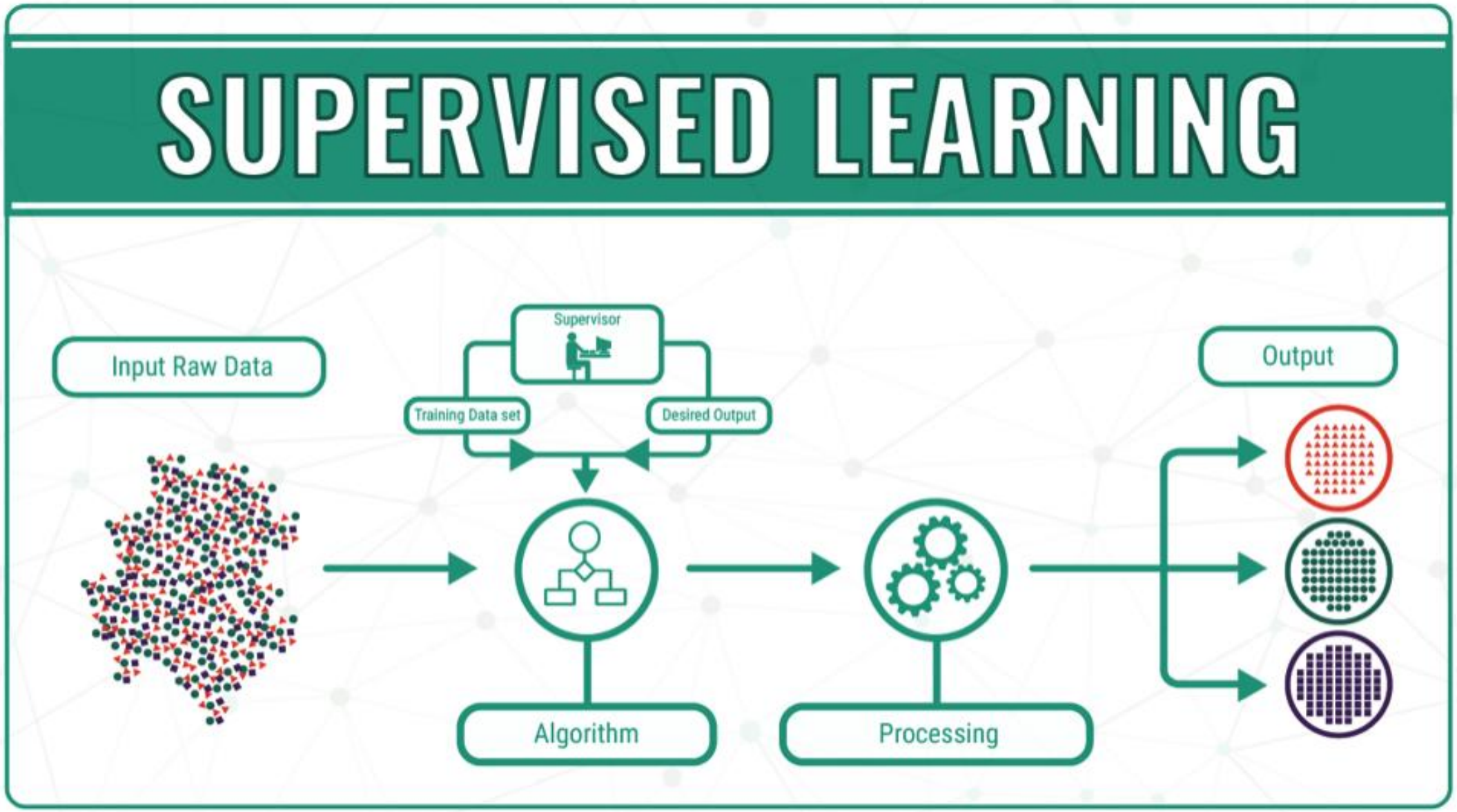
- Uses data to find parameters of a model.
- Model can then be used to make future predictions.
- Can be used to estimate parameters of any type of model.

Taxonomy of Machine Learning





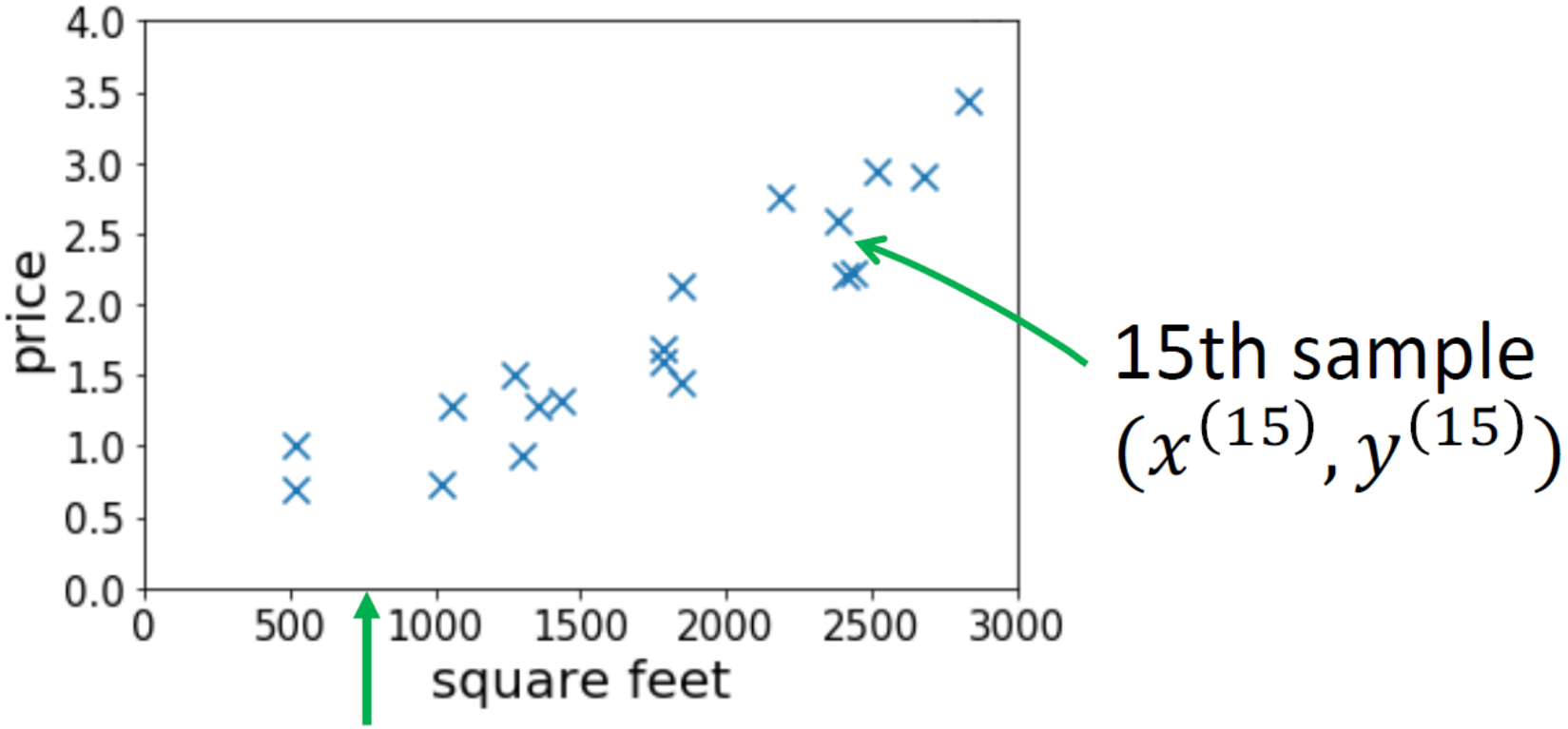
Supervised Learning



Housing Price Prediction

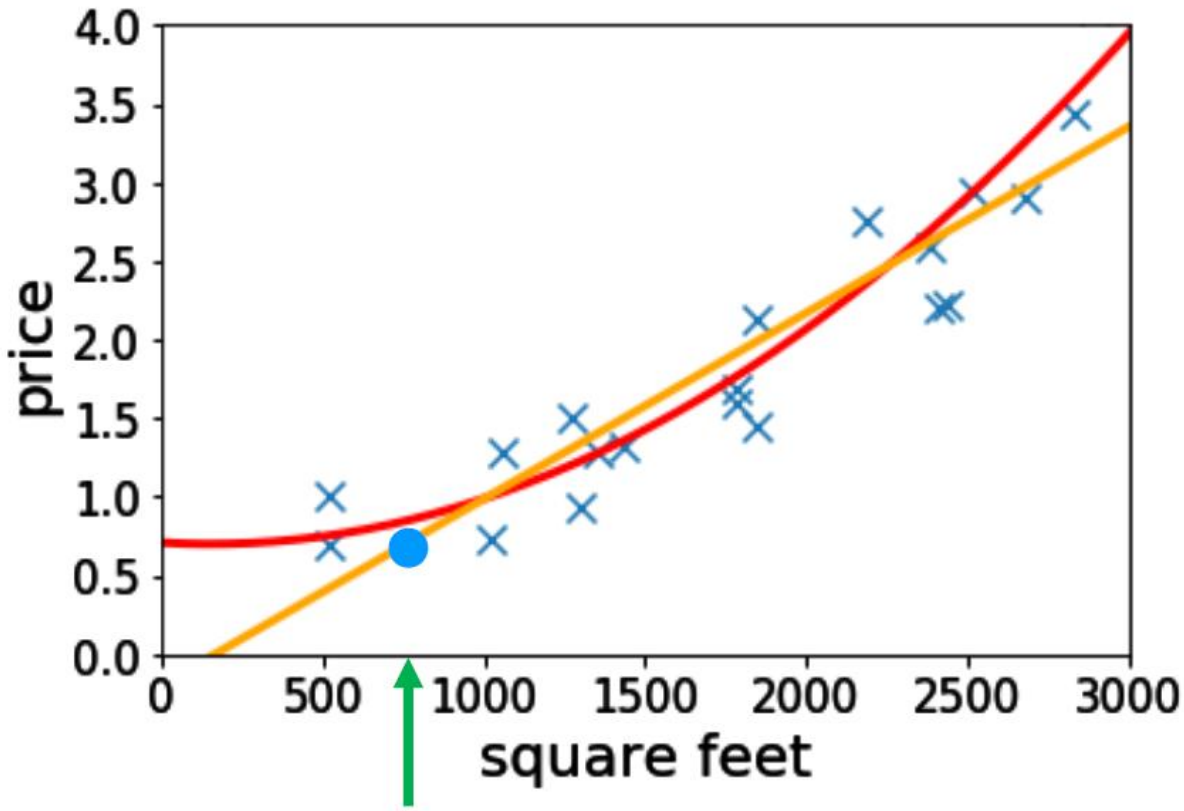
- Given: A dataset of n samples

- Task: If a house has x square feet, predict its price.



Housing Price Prediction

- Given: A dataset of n samples
- Task: If a house has x square feet, predict its price.
- Can have more features



Regression vs Classification

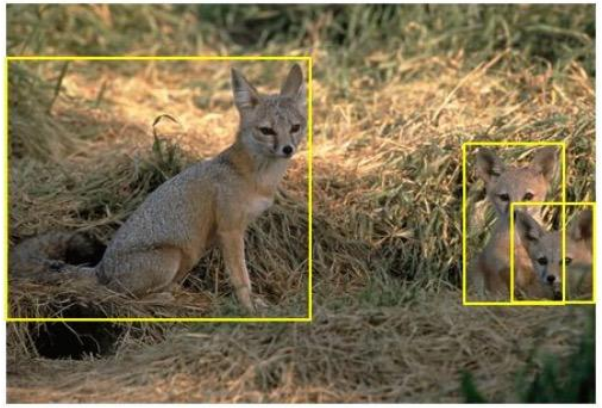
- Regression: Prediction of a continuous variable.
- Classification: Prediction of a discrete variable.

Image Classification

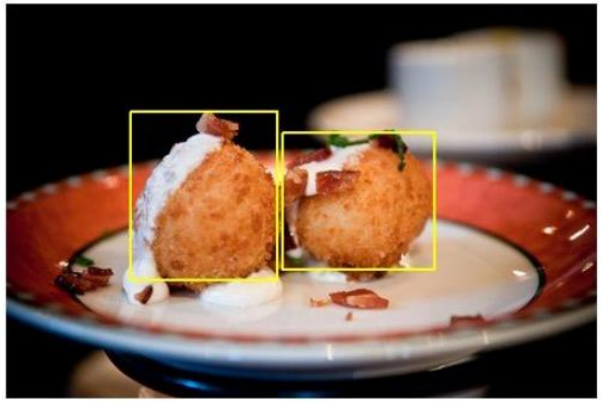
- x = raw pixels of the image, y = the main object

Object Localization and Detection

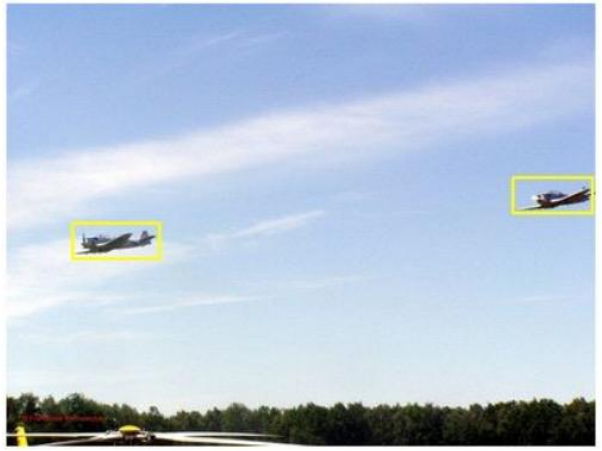
- x = raw pixels of the image, y = the bounding boxes



kit fox



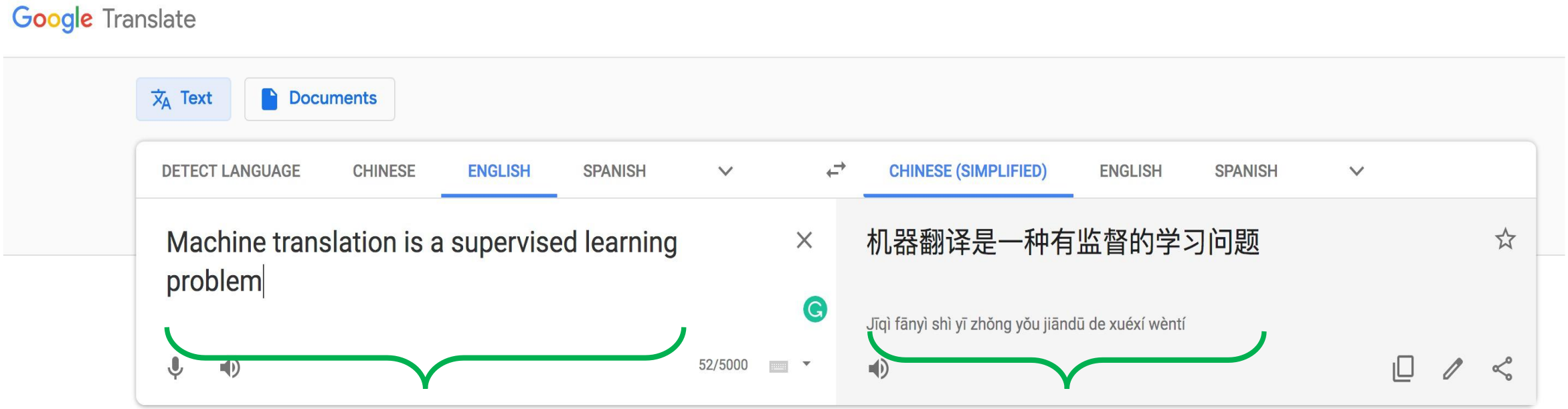
croquette



airplane

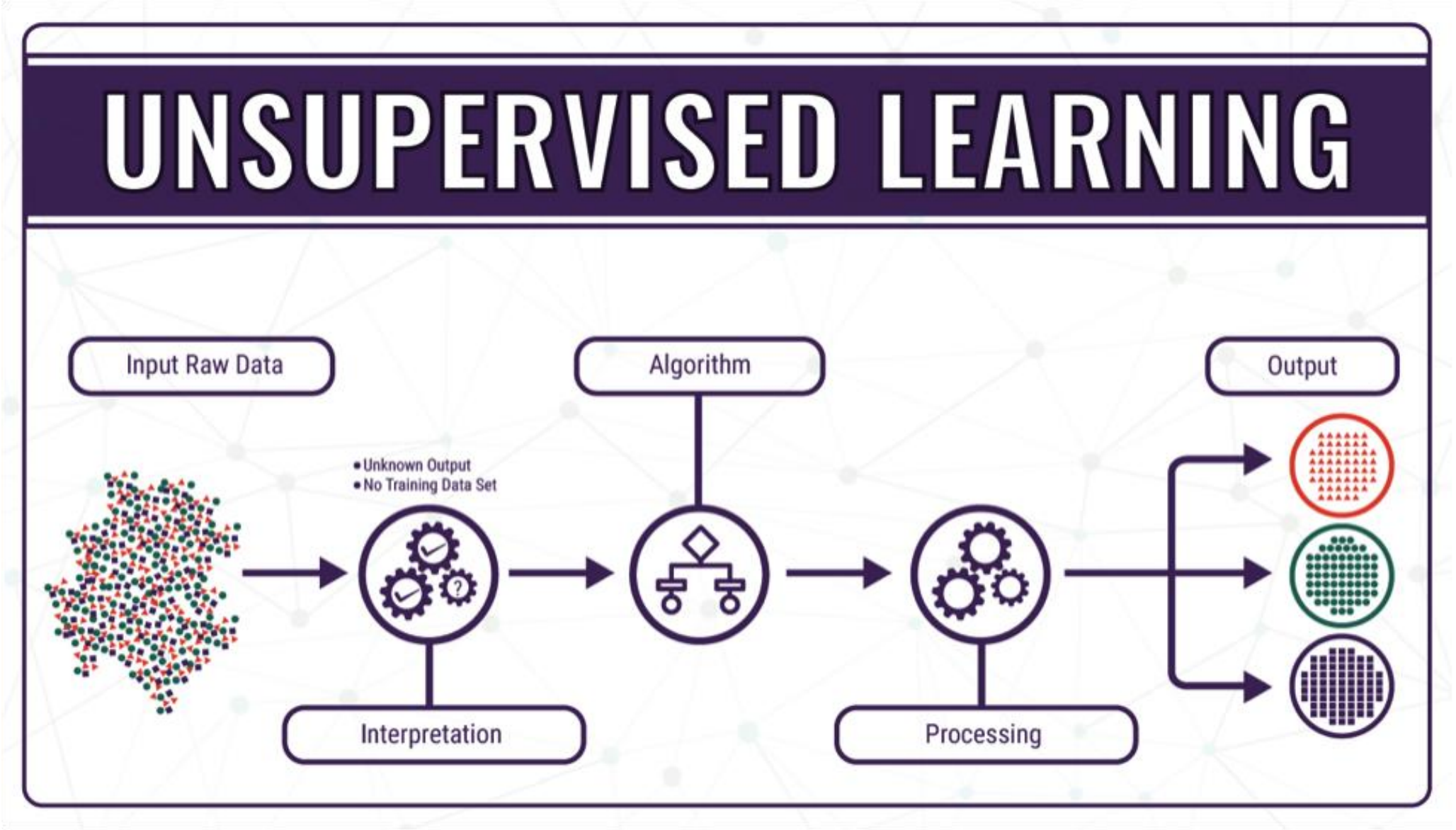
frog

Machine Translation



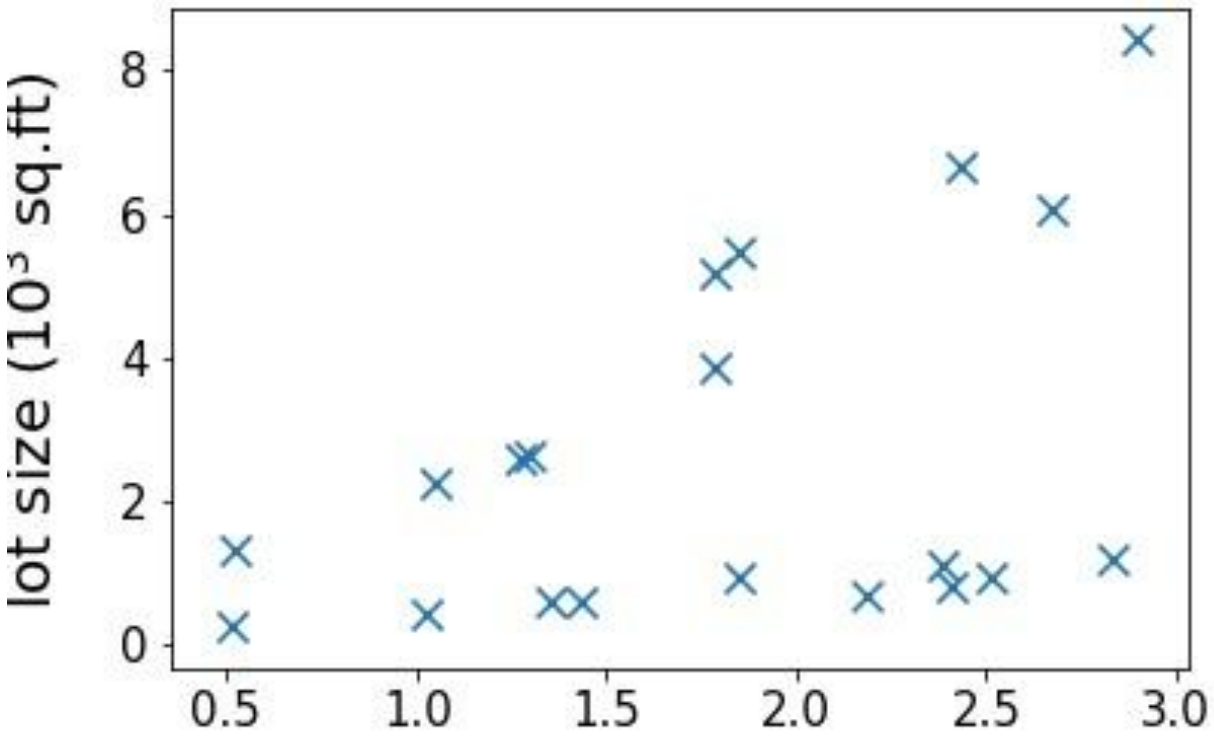
x $\longrightarrow$ y

Unsupervised Learning

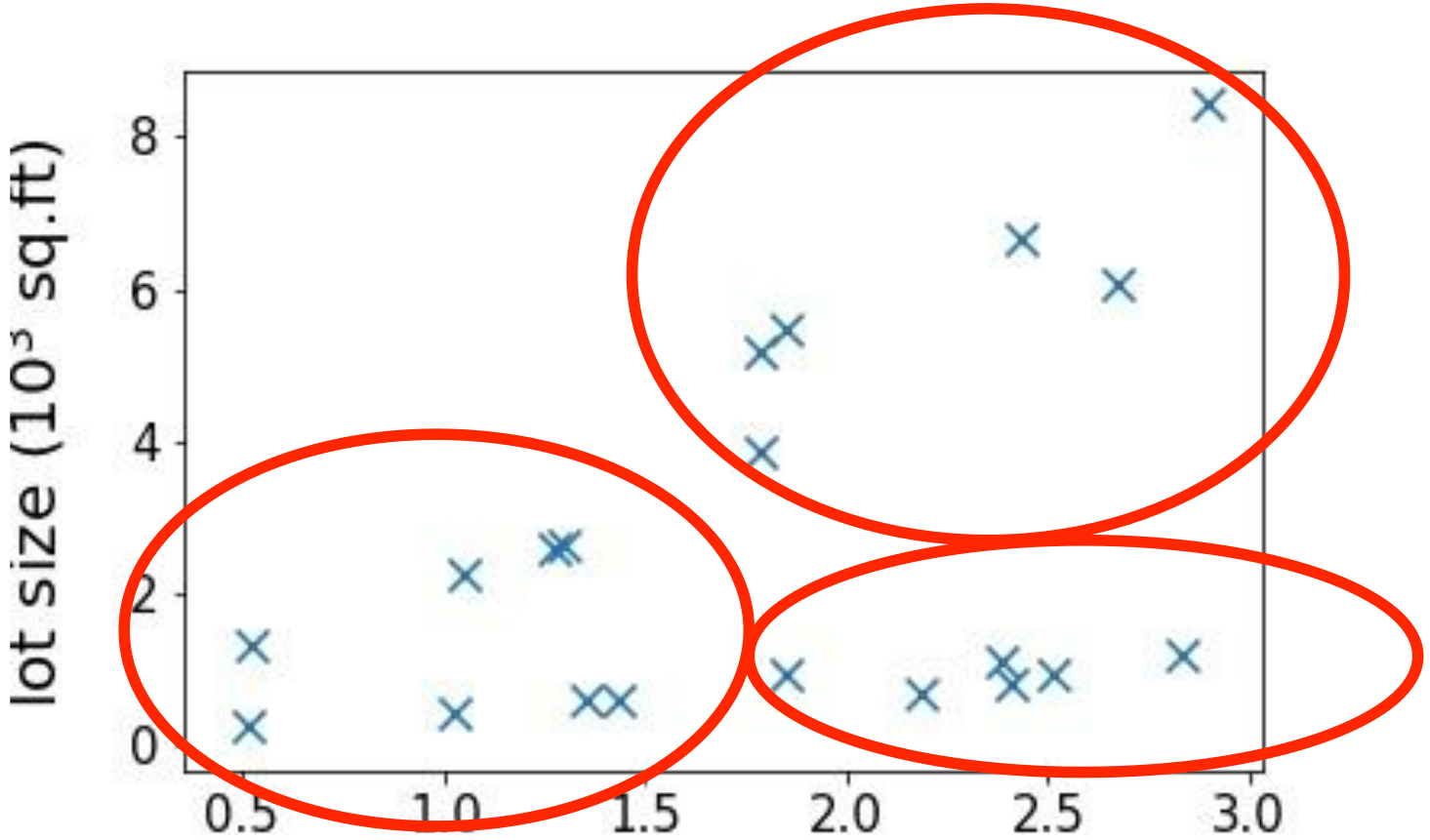


Unsupervised Learning

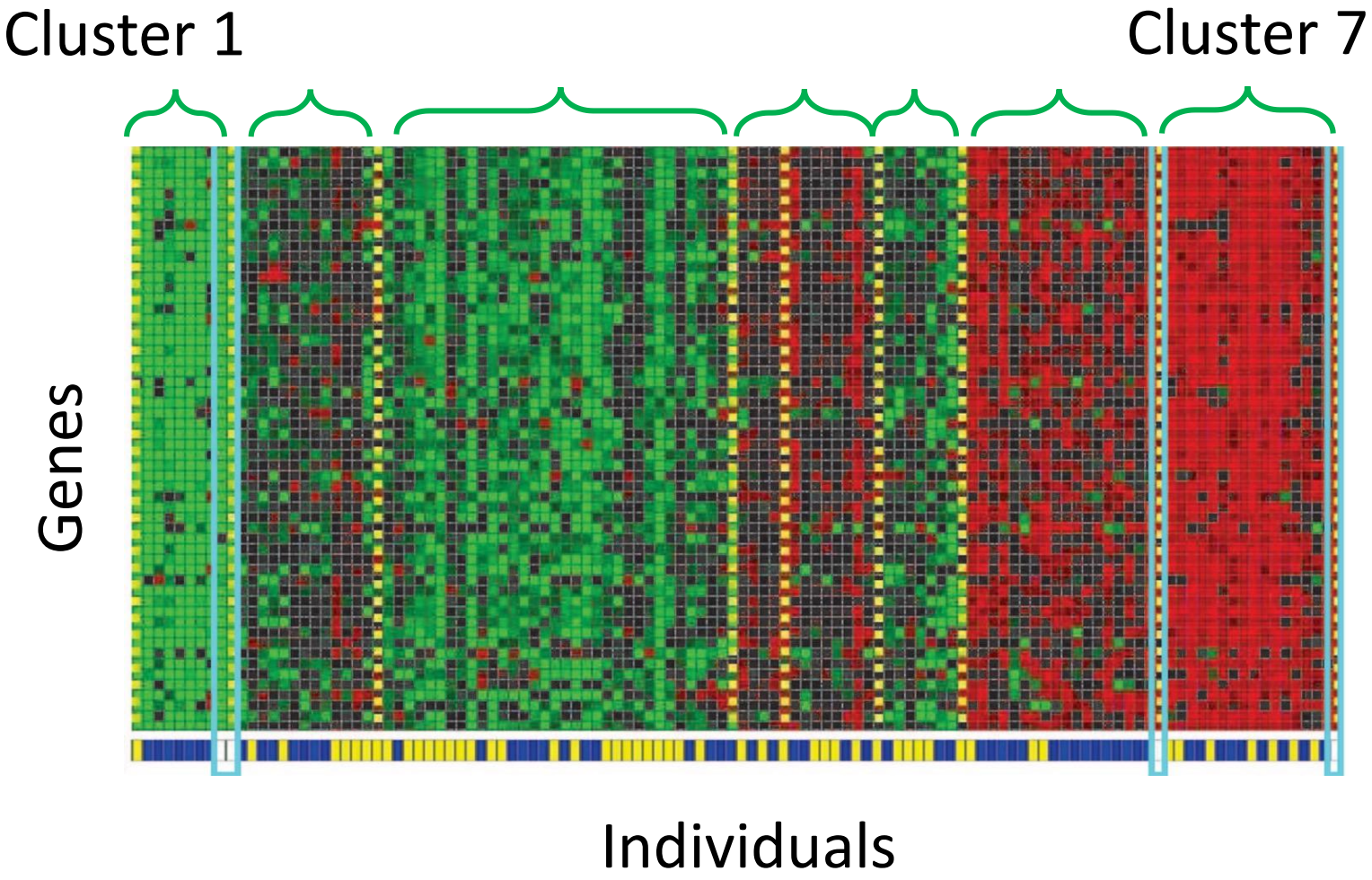
- Dataset contains no labels
- Task: To find structure in the data



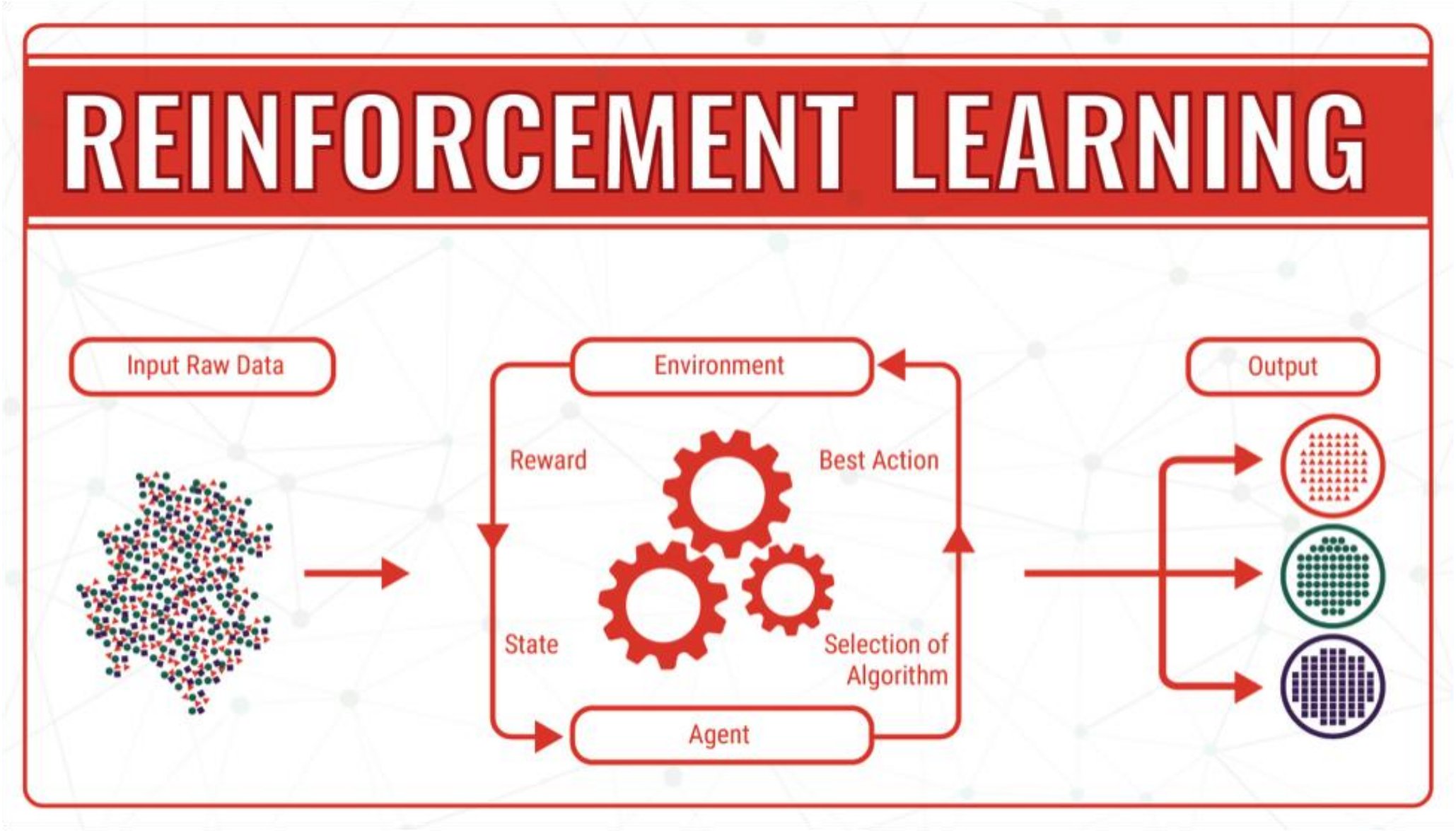
Clustering



Clustering Genes

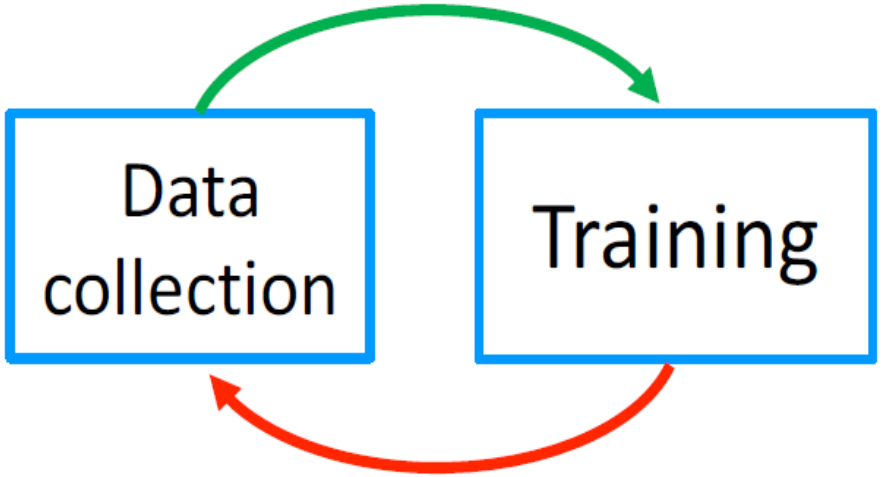


Reinforcement Learning



Reinforcement Learning

Try the strategy and collect feedbacks



Improve the strategy based on the feedbacks

AlphaGo

- AlphaGo is a computer program that plays the board game Go.
- Succeeded by an even more powerful version known as AlphaGo Zero, which was completely self-taught without learning from human games.
- Succeeded by a program known as MuZero which learns without being taught the rules.
- Use reinforcement learning

ML Applications

- ML has moved from the lab to the real world

ML Applications

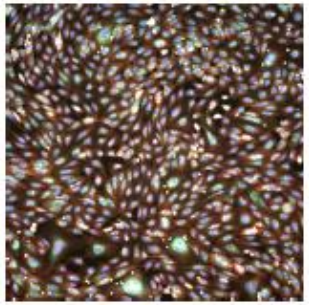
- Machine translation has improved a lot and has started helping people in terms of connecting on a global level
- Autonomous driving has the potential to reduce road accident and congestion.

ML Applications

- ML in healthcare:
  - Interpreting chest x rays

  - Diagnosing diabetic retinopathy

  - Response of covid19 infected cells to certain drugs



ML Applications

- Large Language Models

ML Risks

- Privacy Concerns
- Might lead to violation of privacy
- Security risks
- Bias problem
- Fairness problem
- Creation of fake content
- Energy Consumption

Supervised Learning

- A hypothesis or a prediction function can be represented as:

$$h: \mathcal{X} \rightarrow \mathcal{Y}$$

- x can be an email while y can be spam or not
- x can be house data while y can be price
- Training set: pairs of x and y values
- Given a training set, goal is to produce a good prediction of h.
- Regression: y is continuous
- Classification: y is discrete

Regression Model

$$x \longrightarrow \boxed{h} \longrightarrow y \in \mathbb{R} \text{ response}$$

- Poverty mapping

- Housing prices

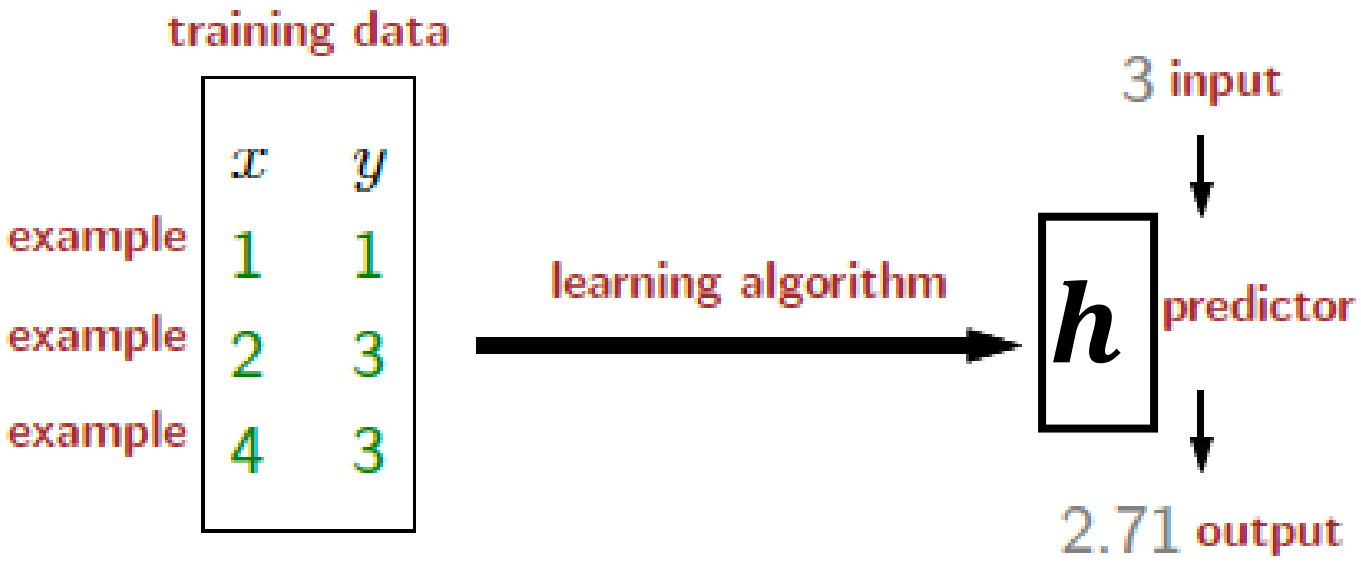
Linear Regression

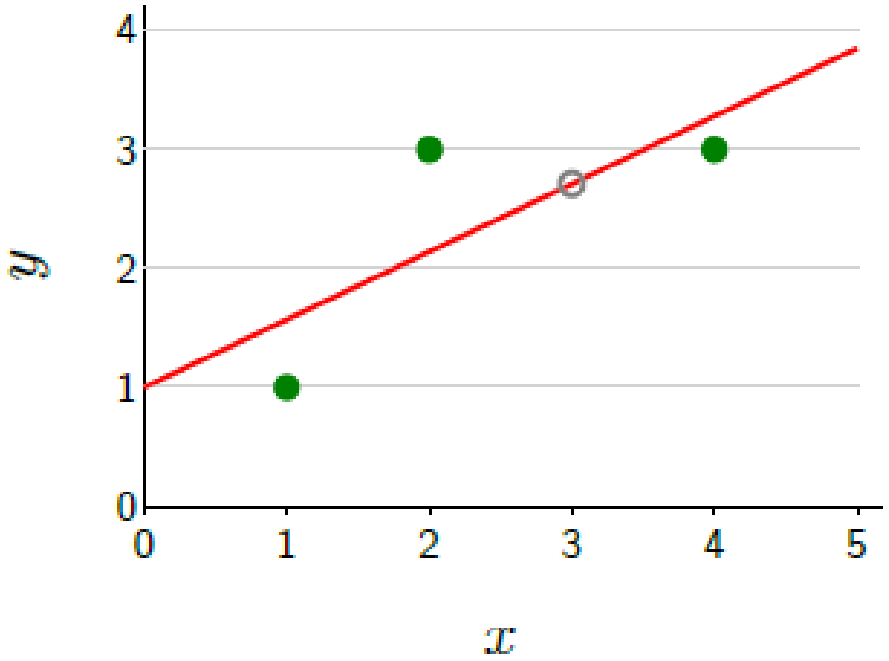
- How do we represent h?

$$h(\boldsymbol{x}) = \theta_0 + \theta_1 x_1 + \theta_2 x_2 + \cdots + \theta_n x_n$$

$$h(\boldsymbol{x}) = \sum_{i=0}^{n} \theta_i x_i$$

Linear Regression





- **Hypothesis class**: The family of possible predictors
- **Loss function**: Determines the goodness of a prediction
- **Optimization algorithm**: Computes the best predictor

Hypothesis Class

- Suppose we consider linear predictors

$$h(\boldsymbol{x}) = \boldsymbol{w}^{\boldsymbol{T}}\phi(\boldsymbol{x}) \qquad (Score)$$

$$weight\ vector: \boldsymbol{w} = [w_0 \ \ w_1 \ \ w_2 \ \dots \ w_n]$$

$$feature\ extractor:\ \phi(\boldsymbol{x}) = [1 \ \ x_1 \ \ x_2 \ \dots \ x_n]\ (feature\ vector)$$

Loss Function

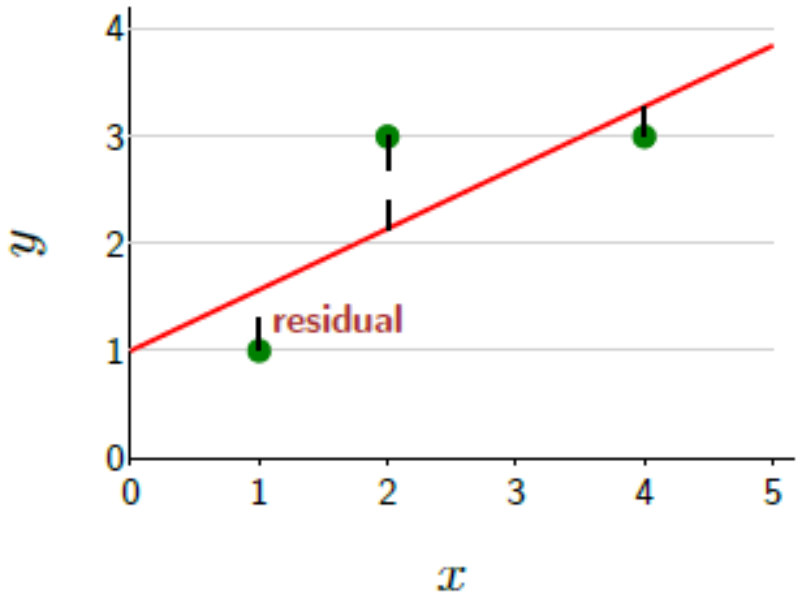
training data $\mathcal{D}_{\text{train}}$

x	y
1	1
2	3
4	3

$$h(\boldsymbol{x}) = \boldsymbol{w}^{\boldsymbol{T}}\phi(\boldsymbol{x})$$

$$\boldsymbol{w} = [1 \quad 0.57]$$

$$\phi(\boldsymbol{x}) = [1 \quad x]$$



- Residual is difference between prediction and actual value

$$Loss(\boldsymbol{x}, y, \boldsymbol{w}) = \frac{1}{2}[h(\boldsymbol{x}) - y]^{\boldsymbol{2}} = \frac{1}{2}[\boldsymbol{w}^{\boldsymbol{T}}\phi(\boldsymbol{x}) - y]^{\boldsymbol{2}} \quad (\boldsymbol{Squared\ Loss})$$

- $TrainLoss(\boldsymbol{w})$ is the average of loss function over all training examples

Finding Gradient

$$TrainLoss(\boldsymbol{w}) = \frac{1}{2N}\sum_{i=1}^{N}[h(\boldsymbol{x_i}) - y_i]^2 = \frac{1}{2N}\sum_{i=1}^{N}[\boldsymbol{w^T}\phi(\boldsymbol{x_i}) - y_i]^2$$

- What is the gradient of the above function?

$$\nabla_{\boldsymbol{w}} TrainLoss(\boldsymbol{w}) = \frac{1}{N}\sum_{i=1}^{N}[\boldsymbol{w^T}\phi(\boldsymbol{x_i}) - y_i]\phi(\boldsymbol{x_i})$$

Optimization Algorithm

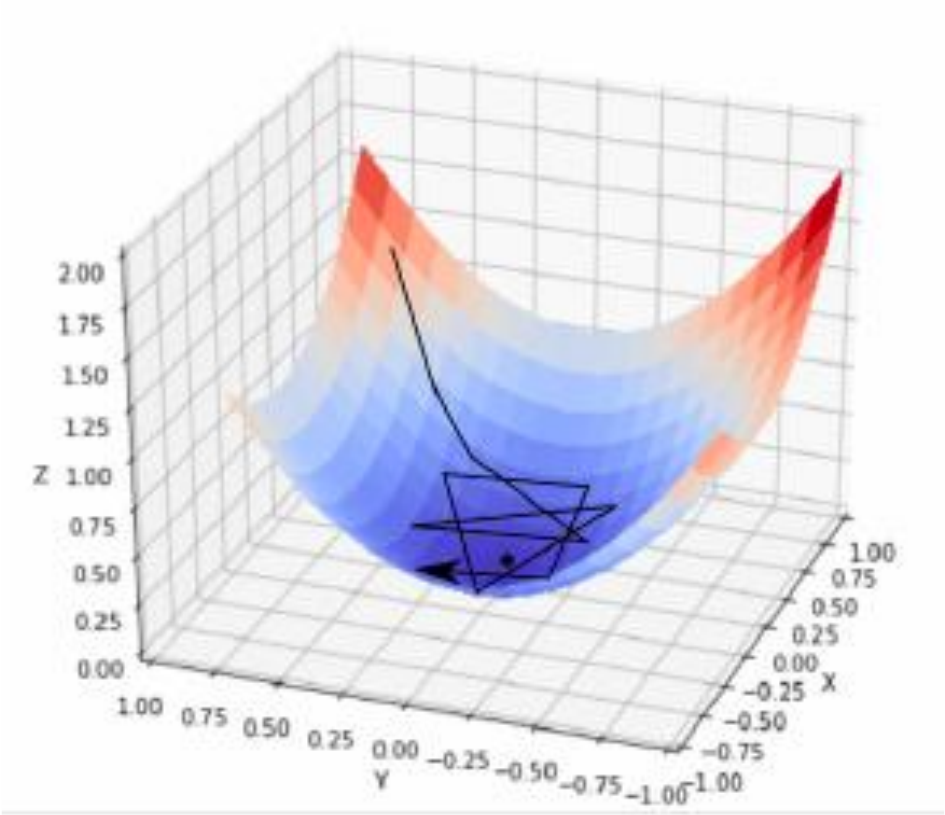
- Find weight vector that minimizes training loss ($TrainLoss(\boldsymbol{w})$)
- Gradient points in which direction?
- Gradient Descent

Algorithm: gradient descent

Initialize $\mathbf{w} = [0, \ldots, 0]$

For $t = 1, \ldots, T$: epochs

$$\mathbf{w} \leftarrow \mathbf{w} - \underbrace{\eta}_{\text{step size}} \underbrace{\nabla_{\mathbf{w}} \text{TrainLoss}(\mathbf{w})}_{\text{gradient}}$$



- Possible problem in gradient descent?

Gradient Descent for Large Datasets

$$TrainLoss(\boldsymbol{w}) = \frac{1}{2N}\sum_{i=1}^{N}[h(\boldsymbol{x_i}) - y_i]^2$$

Algorithm: gradient descent

Initialize $\mathbf{w} = [0, \ldots, 0]$

For $t = 1, \ldots, T$:

$\mathbf{w} \leftarrow \mathbf{w} - \eta \nabla_{\mathbf{w}} \text{TrainLoss}(\mathbf{w})$

- Each iteration of GD goes over all training examples
- GD computationally expensive if dataset is large or number of features is huge

Stochastic Gradient Descent

$$TrainLoss(\boldsymbol{w}) = \frac{1}{2N}\sum_{i=1}^{N}[h(\boldsymbol{x_i}) - y_i]^2$$

Algorithm: stochastic gradient descent

Initialize $\mathbf{w} = [0, \ldots, 0]$
For $t = 1, \ldots, T$:
    For $(x, y) \in \mathcal{D}_{\text{train}}$:
        $\mathbf{w} \leftarrow \mathbf{w} - \eta \nabla_{\mathbf{w}} \text{Loss}(x, y, \mathbf{w})$

- SGD updates weights based on each example
- Can also use minibatch SGD

Questions?